HDOC DAY-73

C Programming Activity

Topic: Enumerations (enum) in C

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Experiment: Enumeration in C

Aim: To understand the use of enumeration (enum) in C by representing days of the week and traffic light states with named integer constants.

Concept / Understanding: An enumeration is a user-defined data type in C that gives meaningful names to integer constants. By default, the first enum constant gets the value 0 and the following constants increase by 1. This makes programs easier to read and understand.

Program 1: Days of the Week

Problem Statement: Create an enumeration for days (SUNDAY to SATURDAY) and print each day with its integer value.

Input: No input

Expected Output: SUNDAY = 0 through SATURDAY = 6.

Algorithm

1. Start the program.
2. Declare an enum named Days containing SUNDAY, MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY and SATURDAY.
3. Use printf() to display each enum constant and its corresponding integer value.
4. Stop the program.

C Program

```
#include <stdio.h>

int main()
{
   enum Days
   {
      SUNDAY, MONDAY, TUESDAY, WEDNESDAY,
      THURSDAY, FRIDAY, SATURDAY
   };

   printf("SUNDAY = %d\n", SUNDAY);
   printf("MONDAY = %d\n", MONDAY);
   printf("TUESDAY = %d\n", TUESDAY);
   printf("WEDNESDAY = %d\n", WEDNESDAY);
   printf("THURSDAY = %d\n", THURSDAY);
   printf("FRIDAY = %d\n", FRIDAY);
   printf("SATURDAY = %d\n", SATURDAY);

   return 0;
}
```

Test / Output Table

Input	Output
No Input	SUNDAY = 0 MONDAY = 1 TUESDAY = 2 WEDNESDAY = 3 THURSDAY = 4 FRIDAY = 5 SATURDAY = 6

Program 2: Traffic Light Enumeration

Problem Statement: Define an enumeration for traffic lights (RED, YELLOW, GREEN) and print 'Stop', 'Wait', or 'Go' based on the selected value.

Given Input: GREEN

Expected Output: Go

Algorithm

5. Start the program.
6. Declare an enum named TrafficLight with RED, YELLOW and GREEN.
7. Store GREEN in an enum variable named light.
8. Use a switch statement to check the value of light.
9. Print Stop for RED, Wait for YELLOW and Go for GREEN.
10. Stop the program.

C Program

```
#include <stdio.h>

int main()
{
   enum TrafficLight
   {
      RED, YELLOW, GREEN
   };

   enum TrafficLight light = GREEN;

   switch (light)
   {
      case RED:
```

```
            printf("Stop");
            break;
        case YELLOW:
            printf("Wait");
            break;
        case GREEN:
            printf("Go");
            break;
        default:
            printf("Invalid Traffic Light");
    }

    return 0;
}
```

Test / Output Table

Input / Selected Value	Output
GREEN	Go
RED	Stop
YELLOW	Wait

Result / Conclusion

Both programs were implemented successfully using enumeration in C. The first program shows the default integer values assigned to enum constants, while the second program uses an enum value with a switch statement to display the appropriate traffic light instruction.